Power BI Project:
Visualisation on Chocolate Sales Dataset 2023-2024

1. Project Objectives

This project analyses a chocolate sales dataset spanning 2023-2024 to uncover actionable insights across three core dimensions: sales performance, product preferences, and customer behaviour.

From a sales perspective, the analysis examines trends in order volumes, revenue, and profitability across different time periods and six key markets (Canada, the United Kingdom, the United States, France, Australia, and Germany) to identify growth opportunities and market disparities.

On the product side, the evaluation assesses the relative popularity of chocolate brands, categories, cocoa intensities, pack weights, and retail channels, with the goal of understanding what drives purchase decisions and where portfolio or distribution optimisation may be warranted.

From a customer standpoint, segmentation analysis by age, gender, and loyalty status is conducted to build a clearer picture of the target audience and their purchasing patterns, including acquisition trends and repeat-buying behaviour.

Collectively, these insights are intended to equip businesses with the evidence base needed to make informed strategic decisions - whether in marketing, product development, channel investment, or geographic expansion - and to strengthen their competitive position in the premium chocolate market.

2. About the Dataset

A. Data Source:

The dataset used in this project is a synthetic retail transaction record covering chocolate sales across multiple stores, customers, and products over the 2023–2024 period. It was sourced from Kaggle and is publicly available at the link below.

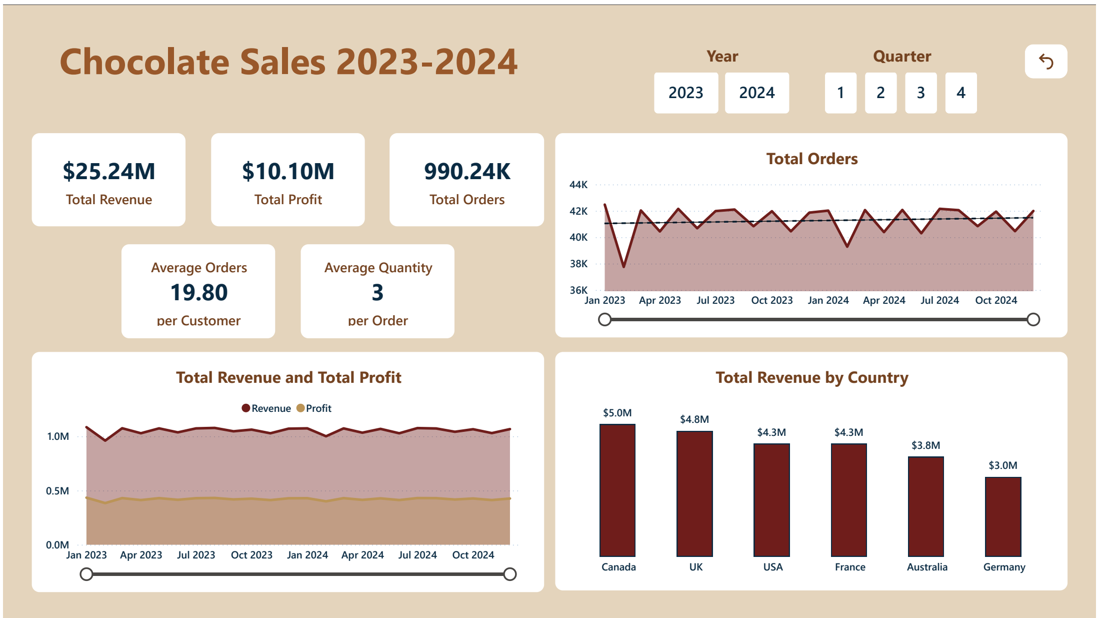
Link: kaggle.com/datasets/ssssws/chocolate-sales-dataset-2023-2024

B. Data Structure:

The dataset follows a **Star Schema** design, comprising one central fact table surrounded by four supporting dimension tables:

- **Sales** *(Fact)*: The core of the schema, capturing transaction-level detail including Order ID, Order Date, Quantity, Unit Price, and related sales metrics.
- **Products** *(Dim)*: Descriptive attributes of each chocolate product, such as Category, Cocoa Percentage, and Pack Weight.
- **Stores** *(Dim)*: Contextual information about retail locations, including Store Type and Country.
- **Customers** *(Dim)*: Demographic and behavioural profiles of each customer, covering Customer ID, Age, Gender, and Loyalty Status.
- **Calendar** *(Dim)*: A date table that enables time-based analysis across Day, Month, and Year.

3. Dashboards & Key Insights:



A. Dashboard 1:

I. Overall Performance

- Total revenue across both years reached **$25.24M**, with total profit of **$10.10M**, representing a **~40% profit margin** remarkably consistent across both years.
- Total orders across both years came to **990.24k**, split almost evenly - **494.70k in 2023** and **495.54k in 2024** - indicating a stable, mature business with minimal year-over-year growth.

II. Orders Trend

- Monthly orders ranged between **40k and 43k**, with occasional dips toward the **36k–38k** range visible at certain points in the time-series charts.
- Despite being the most commercially associated month with chocolate, **orders appear to soften in early months** (Q1 dips are visible in both years), suggesting Valentine's Day demand may not translate into a notable spike at the aggregate level - or may even correlate with a brief lull as retailers restock.
- Order volumes appear broadly **flat year-on-year**, with no clear acceleration or seasonal peak that dominates the other quarters.

III. Revenue & Profit

- Revenue was evenly split, **$12.61M in 2023** vs **$12.63M in 2024,** with a marginal increase of just ~$20k (+0.16%).
- Profit mirrored this closely with **$5.04M (2023)** vs **$5.05M (2024)**, indicating that cost structures remained stable and pricing power was maintained.
- The **40% profit margin** held solid across both years - a sign of strong operational discipline and consistent product mix.
- Monthly revenue and profit charts show a relatively **smooth, stable pattern** without dramatic spikes, though minor undulations suggest some modest seasonality within each year.

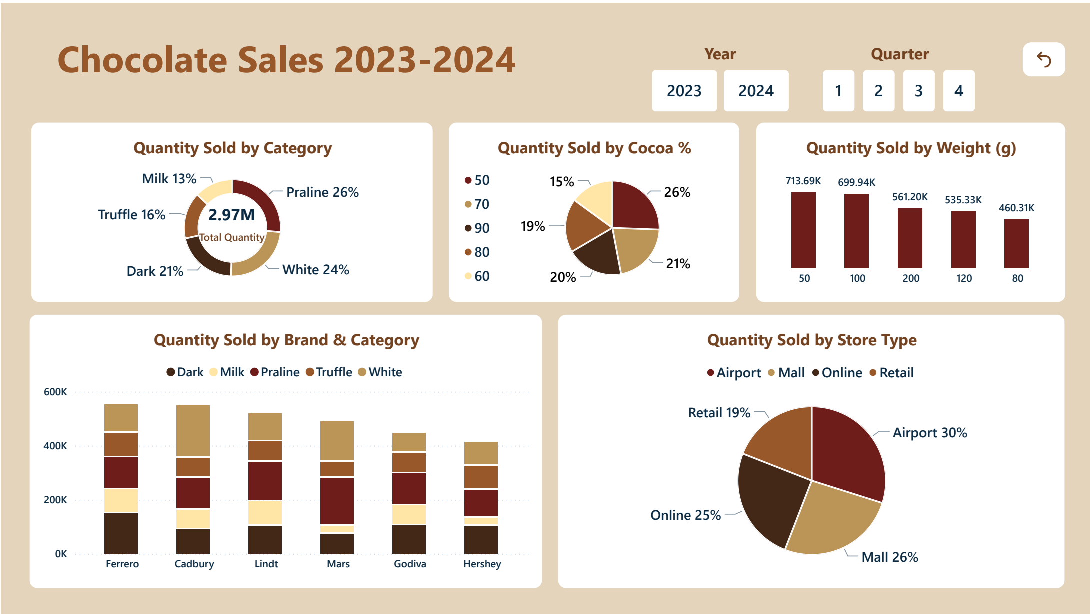
IV. Geographic Performance

- **Canada** is the leading market at **$5.0M** in total revenue, followed closely by the **UK at $4.8M**.
- The **USA and France** are neck-and-neck at **~$4.3M each**, forming a solid mid-tier.
- **Australia ($3.8M)** and **Germany ($3.0M)** round out the bottom.

- One interesting geographic shift was in **2024,** when **France overtook the USA** ($2.2M vs $2.1M), suggesting France is the faster-growing market of the two, worth monitoring closely.
- The spread between the top (Canada) and bottom (Germany) markets was significant as Canada generated **~67% more revenue than Germany**, indicating either a much larger customer base or higher spend per customer.

V. Customer Behaviour

- On average, each customer placed **~19.80 orders over the full two-year period** (~9.89/ year in 2023, ~9.91/year in 2024) - remarkably consistent and suggesting strong **repeat purchase behaviour and customer loyalty**.
- Each order contained an average of **3 items**, which held constant across both years and all markets - pointing to a predictable and stable basket size.
- The consistency in both order frequency and basket size suggests the customer base is habitual rather than opportunistic - these are **loyal, recurring buyers** rather than impulse purchasers.



B. Dashboard 2:

I. Category Preferences

- **Praline dominated** at 26% of quantity sold, followed by **White (24%)** and **Dark (21%)**, making these three the clear top tier - together accounting for **71% of all units sold**.
- **Truffle (16%)** and **Milk (13%)** lagged behind, with Milk being the weakest performer despite being arguably the most universally recognised chocolate type. This suggests the customer base skews toward **premium and artisan varieties** over everyday confectionery.
- The category split was virtually **identical between 2023 and 2024** (Truffle nudged down 1% to 15% in 2024), reinforcing that consumer preferences in this market are highly stable and brand/category loyalty is strong.

II. Cocoa Percentage

- The **50% cocoa** segment was the most popular at **26%**, followed by **70% (21%)**, **90% (20%)**, **80% (19%)**, and **60% (15%)**.
- It's notable that the two *most intense* cocoa options - **90% and 80%** — together account for **39%** of quantity sold, pointing to a meaningfully large segment of **dark chocolate enthusiasts** willing to go beyond the mainstream.

- The relative unpopularity of **60% cocoa** (the lowest share at **15%**) is somewhat counterintuitive, as it sits in the "middle ground" - perhaps perceived as neither indulgent enough nor distinctive enough compared to the extremes.

III. Pack Weight

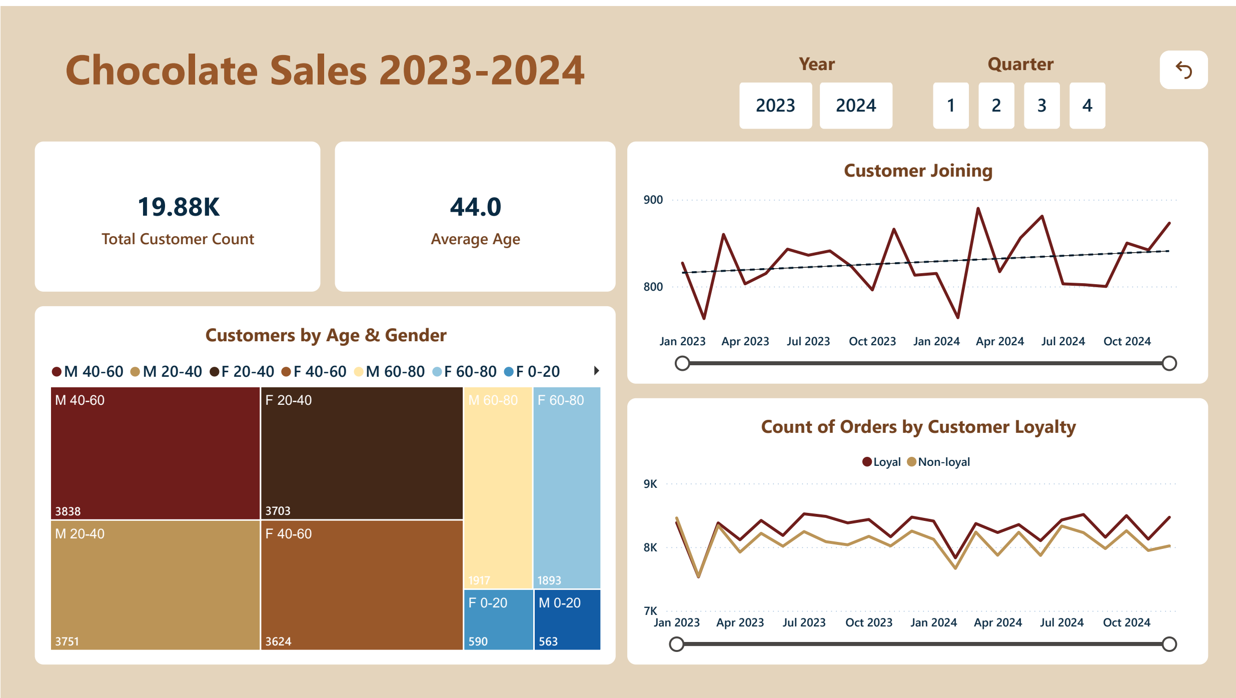
- The **50g pack was the top seller at 713.69k units**, followed closely by the **100g pack at 699.94k**, together making up nearly half of all units sold.
- Sales declined progressively with weight: **200g (561.20K) → 120g (535.33K) → 80g (460.31K)**, suggesting customers prefer either a **small single-serve or a standard sharing size**, with the mid-range weights (80g, 120g) being less compelling.
- The 80g pack was the weakest performer despite sitting between two stronger sellers (50g and 100g) - it may be a candidate for **rationalisation or repositioning**.
- Year-on-year, weight preferences were almost perfectly stable, with no meaningful shift between 2023 and 2024.

IV. Sales Channels

- **Airports led all channels at ~30% share (~891k units)**, a striking result that likely reflects impulse gifting, travel retail premiumisation, and a captive high-spending audience.
- **Malls (26%, ~773k)** and **Online (25%, ~743k)** were closely matched, suggesting the business has a healthy omnichannel presence between physical retail and e-commerce.
- **Retail stores were the weakest at 19% (~566k units)** - notably behind even online, which may reflect fewer locations, lower foot traffic, or a less premium in-store experience versus airport and mall environments.
- Channel mix was **identical across 2023 and 2024**, suggesting no channel shift is underway - though the near-parity of online and mall is worth watching as e-commerce trends typically favour online growth over time.

V. Brand & Category Performance

- **Ferrero was the clear #1 brand** by quantity sold, with a strong lead over the rest of the field - its strength appears broad across multiple categories.
- The ranking of **Cadbury → Lindt → Mars → Godiva → Hershey** followed, with **Hershey being the weakest performer** even though it is one of the world's largest chocolate brands, possibly reflecting a mismatch between Hershey's mass-market positioning and this dataset's more premium-skewing customer base.
- Across brands, **Praline and White** categories consistently drove high volumes, aligning with the overall category mix.
- Brand rankings and category contributions were **consistent year-on-year**, with no brand showing a notable surge or decline between 2023 and 2024.



C. Dashboard 3:

I. Overall Customer Base

- The business served a total of **19.88K customers** across the two-year period, with an **average age of 44** - firmly placing the core customer in middle adulthood and aligning well with the premium, treat-oriented product profile identified in the product analysis.
- New customer acquisition was **consistent and steady**, with monthly joiners tracking in the **800–900 range** throughout most of the period, with occasional dips toward 800 and 750 during February in both years. There is no visible acceleration or deceleration, reflecting the same plateau dynamic seen in orders and revenue.

II. Gender Split

- The gender balance was remarkably even: **10,069 males (50.6%)** vs **9,810 females (49.4%)**.
- This parity held across age groups too, with male and female counts closely mirroring each other at every age group - suggesting the product has **truly universal appeal** with no meaningful gender skew to exploit or address in marketing.

III. Age Distribution

- The **40-60** age bracket was the largest segment at 7,462 customers, closely followed by **20-40** at 7,454, virtually tied and together accounting for ~75% of the entire customer base.
- The **60–80 group (3,810)** came third, representing about 19% - a meaningfully large senior segment that likely skews toward gifting and self-indulgence purchases.
- The **0–20 group was the smallest at just 1,153 customers (~5.8%)** - predictably low given purchasing power constraints, but also a potential future growth cohort as these customers age into the 20-40 bracket.
- The average age of **44** is consistent with the 40–60 bracket being the core — this is a **Gen X and older Millennial-dominated** customer base.

IV. Gender & Age Breakdown

- Breaking down by both dimensions reveals a consistent pattern where **males slightly outnumber females in every age group**:
  - 40-60: M 3,838 vs F 3,624
  - 20-40: M 3,751 vs F 3,703
  - 60-80: M 1,917 vs F 1,893
  - 0-20: M 590 vs F 563
- The male lead is marginal in every bracket, never exceeding ~6% within any group, confirming that **gender is not a meaningful differentiator** in this customer base.
- The near-symmetry across all four age bands suggests the business has not accidentally skewed its acquisition toward any particular demographic, and its marketing and channel mix are working evenly across groups.

V. Customer Loyalty

- Orders from loyal and non-loyal customers tracked closely throughout the entire period, both hovering in the 7K–9K orders per month range with no dramatic divergence.
- Neither group showed a meaningful trend upward or downward - loyalty levels appear **structurally stable** rather than improving or deteriorating.
- The proximity of loyal vs non-loyal order volumes is interesting. It could indicate that the business's loyalty programme is **not strongly differentiating behaviour**, or that non-loyal customers are simply high-frequency repeat buyers who haven't formally enrolled.